

Thermal Field Theory and Open Quantum Systems

*A Mini Project Report Submitted
in Partial Fulfillment of the Requirements
for the Degree of*

Bachelor of Technology

by

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CERTIFICATE

*This is to certify that the work contained in this thesis entitled “**Thermal Field Theory and Open Quantum Systems**” is a bonafide work of **Debraj Das** (Roll No. **230121016**), carried out in the Department of Physics, Indian Institute of Technology Guwahati under my supervision and that it has not been submitted elsewhere for a degree.*

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April, 2026

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Abstract

In this report, we study the behavior of quantum fields at finite temperature and their interaction with external environments. We begin by developing the formalism of thermal field theory, focusing on the path integral representation, thermal Green's functions, and the role of the spectral function. Both imaginary-time (Matsubara) and real-time approaches are discussed to describe equilibrium and dynamical properties.

We then extend the analysis to open quantum systems, where a system interacts with an environment. Using the influence functional formalism, we derive an effective description of the system by integrating out environmental degrees of freedom. This leads to nonlocal dynamics characterized by dissipation and noise, with their structure determined by the thermal properties of the environment. The results illustrate how thermal field theory naturally connects to the description of open quantum systems.

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Chapter 1

Introduction

Many physical systems of interest, ranging from condensed matter systems to high-energy physics and cosmology, exist at finite temperature. Moreover these systems interact with surrounding environments. We also assume that the system we are dealing with is in equilibrium but that is rarely the case. Thus it is necessary to take into account this thermal aspect to accurately determine the real world behavior of these systems.

Quantum Field Theory which is considered the most successful theory of physics is formulated in the zero temperature regime and do not include temperature. This gives rise to finite temperature field theory which connects concepts of statistical ensembles to quantum fields thus allowing us to model these fields keeping temperature in mind.

While Thermal Field Theory incorporates temperature into the formalism, it assumes the system to be in thermal equilibrium and isolated. But real world processes are not isolated and interact with their surroundings which affects their dynamics. So the formalism must be extended to systems that are open and free to interact with their surroundings.

1.1 Organization of The Report

In this work we look into how quantum fields behave when they are free to interact with their surroundings which is kept at a finite temperature.

We start by introducing concepts from thermal field theory from the first principles and study the imaginary time formalism also called the Matsubara formalism and then the imaginary time or thermal propagator. We also derive the partition function of a free scalar field.

Next we discuss the real time formalism since all real world dynamics take place in real time rather than imaginary time. We derive the matrix structure of the thermal propagator for a free field.

Finally we allow the field to interact with its surrounding. We derive the non-local dissipation and noise kernel using the influence functional formalism for the case of two scalar fields coupled together where one of the fields is the system we are interested in whereas the other behaves as the environment. We then keep the environment in thermal equilibrium and find the form of the dissipation and noise kernels.

Chapter 2

Thermal Field Theory

2.1 Introduction

In this chapter we look into the thermalisation of quantum fields in equilibrium with a bath at temperature T . We start with quantum dynamics with single degree of freedom by writing the partition function and propagators using path integral formulation. Then we move onto quantum fields by generalising the results to infinite degrees of freedom by integrating over space. [4]

2.2 Path Integral Formulation

The statistical and thermal properties of a system are encoded in the partition function, which is given by

$$Z(\beta) = \text{Tr} \left(e^{-\beta \hat{H}} \right) \quad (2.1)$$

where $\hat{H}$ is the Hamiltonian operator of the system and $\beta = (k_B T)^{-1}$. Expanding the trace in the position basis we get

$$Z(\beta) = \int dq \langle q | e^{-\beta \hat{H}} | q \rangle \quad (2.2)$$

We notice that the term $e^{-\beta\hat{H}}$ in the above expression has the same form as the evolution operator $e^{-i\hat{H}t}$ but the time variable is now imaginary.

$$t = -i\beta \quad (2.3)$$

Thus if we write $\tau = -i\beta$, then the partition function becomes

$$Z(\beta) = \int dq \langle q | e^{-i\hat{H}(\tau' - \tau)} | q \rangle \quad (2.4)$$

This is called imaginary time formalism as the exponent represent an evolution along imaginary time as β is real [4]. The partition function can now be written as a path integral [4] the end points are also to be integrated. To do this we divide the time evolution into n steps:

$$e^{-\hat{H}(\tau' - \tau)} = \left(e^{-\epsilon\hat{H}} \right)^n, \quad \epsilon = \frac{(\tau' - \tau)}{n} \quad (2.5)$$

Since the Hamiltonian is give by $\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{q})$ we have

$$e^{-\hat{H}(\tau' - \tau)} = \left(e^{-\epsilon(\frac{\hat{p}^2}{2m} + V(\hat{q}))} \right)^n \quad (2.6)$$

In the limit n tends to infinity, according to Trotters formula which states

$$\lim_{n \rightarrow \infty} \left(e^{\frac{A}{n}} e^{\frac{B}{n}} \right)^n = e^{A+B} \quad (2.7)$$

We get

$$\left(e^{-\epsilon(\frac{\hat{p}^2}{2m} + V(\hat{q}))} \right)^n = \prod_{i=1}^n e^{-\epsilon\frac{V(\hat{q})}{2}} e^{-\epsilon\frac{\hat{p}^2}{2m}} e^{-\epsilon\frac{V(\hat{q})}{2}} \quad (2.8)$$

Substituting in equation (2.4) we get

$$Z(\beta) = \int dq \langle q | \prod_{i=1}^n e^{-\epsilon\frac{V(\hat{q})}{2}} e^{-\epsilon\frac{\hat{p}^2}{2m}} e^{-\epsilon\frac{V(\hat{q})}{2}} | q \rangle \quad (2.9)$$

By inserting complete sets of position eigenstates at times $\tau_1, \tau_2, \dots, \tau_n$ the partition function becomes [4]

$$Z(\beta) = \int \prod_{i=0}^n dq_i \langle q_{i+1} | e^{-\epsilon \frac{V(\hat{q})}{2}} e^{-\epsilon \frac{\hat{p}^2}{2m}} e^{-\epsilon \frac{V(\hat{q})}{2}} | q_i \rangle \quad (2.10)$$

$$\langle q_{i+1} | e^{-\epsilon \frac{V(\hat{q})}{2}} e^{-\epsilon \frac{\hat{p}^2}{2m}} e^{-\epsilon \frac{V(\hat{q})}{2}} | q_i \rangle = e^{\frac{V(q_{i+1}) + V(q_i)}{2}} \langle q_{i+1} | e^{-\epsilon \frac{\hat{p}^2}{2m}} | q_i \rangle \quad (2.11)$$

We then insert a complete set of momentum eigenstates to get

$$\langle q_{i+1} | e^{-\epsilon \frac{\hat{p}^2}{2m}} | q_i \rangle = \int \frac{dp_i}{2\pi} e^{ip_i(q_{i+1} - q_i)} e^{-\epsilon \frac{p_i^2}{2m}} \quad (2.12)$$

This expression is in the standard gaussian integral form. On evaluating the gaussian integration:

$$\langle q_{i+1} | e^{-\epsilon \frac{\hat{p}^2}{2m}} | q_i \rangle = \left(\frac{m}{2\pi\epsilon} \right)^{1/2} \exp \left[-\frac{m(q_{i+1} - q_i)^2}{2\epsilon} \right] \quad (2.13)$$

Thus, putting all together we get the expression

$$Z(\beta) = \lim_{\epsilon \rightarrow 0} \left(\frac{m}{2\pi\epsilon} \right)^{n/2} \int \prod_{i=1}^n dq_i \exp \left[-\epsilon \sum_{i=0}^{n-1} \left(\frac{m}{2} \left(\frac{q_{i+1} - q_i}{\epsilon} \right)^2 + V \left(\frac{q_{i+1} + q_i}{2} \right) \right) \right] \quad (2.14)$$

Where we have written $\frac{V(q_{i+1}) + V(q_i)}{2}$ as $V\left(\frac{q_{i+1} + q_i}{2}\right)$ because since the divisions are small, the potential function between q_i and q_{i+1} can be approximated as a straight line and the average of the potential at the end points can be replaced by the value at the middle in accordance to mean value theorem. As ϵ tends to zero the the term in the exponent is nothing but the Riemann sum of the integral

$$S_E[q] = \int_0^\beta d\tau \left[\frac{1}{2} m \dot{q}^2(\tau) + V(q(\tau)) \right] \quad (2.15)$$

Where $S_E[q]$ is called the Euclidean action. Thus the partition function becomes

$$Z(\beta) = \int \mathcal{D}q(\tau) e^{-S_E[q]} \quad (2.16)$$

where one integrates over all paths satisfying

$$q(\beta) = q(0) \quad (2.17)$$

i.e. all paths periodic in imaginary time with a period β . This arises from the KMS condition obtained from the thermal average of the product of two operators as[3].

The thermal average of operators is defined

$$\langle A(t)B(t') \rangle_\beta = \frac{1}{Z(\beta)} \text{Tr} \left(e^{-\beta \hat{H}} A(t) B(t') \right) \quad (2.18)$$

where $A(t)$ and $B(t)$ are two operators in Heisenberg picture.

$$\langle A(t)B(t') \rangle_\beta = \frac{1}{Z(\beta)} \text{Tr} \left(e^{-\beta \hat{H}} A(t) e^{\beta \hat{H}} e^{-\beta \hat{H}} B(t') \right) \quad (2.19)$$

Since $e^{-\beta \hat{H}}$ is an evolution operator in imaginary time evolution and using the cyclicity of trace, we get

$$\langle A(t)B(t') \rangle_\beta = \frac{1}{Z(\beta)} \text{Tr} \left(e^{-\beta \hat{H}} B(t') A(t + i\beta) \right) \quad (2.20)$$

Thus,

$$\langle A(t)B(t') \rangle_\beta = \langle B(t')A(t + i\beta) \rangle_\beta \quad (2.21)$$

Replacing A and B with $\hat{q}(\tau)$:

$$\langle \hat{q}(\tau)\hat{q}(\tau') \rangle_\beta = \langle \hat{q}(\tau')\hat{q}(\tau + \beta) \rangle_\beta \quad (2.22)$$

Thus, $\hat{q}$ is periodic in imaginary time with period β . Due to this periodic nature we can

write

$$\langle T(\hat{q}(-i\beta)\hat{q}(-i\tau)) \rangle = \langle T(\hat{q}(0)\hat{q}(-i\tau)) \rangle \quad (2.23)$$

This periodicity can be interpreted topologically where the imaginary time axis is folded into a circle with circumference β and the operator $\hat{q}$ lives on this circle. It is useful to define the periodic function which will be used to define the propagator later [4]

$$\Delta(\tau) = \langle Tq(-i\tau)q(0) \rangle_\beta \quad (2.24)$$

which follows the property, where τ is in the range $[0, \beta]$

$$\Delta(\tau + \beta) = \Delta(\tau) \quad (2.25)$$

2.3 Real Time Propagator and Spectral Function

Till now we have been working with imaginary time; to model real world phenomenon we need to extend the theory to real time. To do so we define the two point functions

$$D^>(t, t') = \langle \hat{q}(t)\hat{q}(t') \rangle_\beta \quad (2.26)$$

$$D^<(t, t') = \langle \hat{q}(t')\hat{q}(t) \rangle_\beta = D^>(t', t) \quad (2.27)$$

The small β in the subscript denotes that expectation is taken by thermal average. From the periodicity of $\hat{q}$ (KMS condition) we get [4]

$$D^<(t, t') = D^>(t + i\beta, t') \quad (2.28)$$

when $\tau \in [0, \beta]$. We note from eq (2.24) and eq (2.26)

$$\Delta(\tau) = D^>(-i\tau, 0) \quad (2.29)$$

Therefore, the time ordered propagator in real time can be written using the definitions above as

$$D(t, t') = \langle T(\hat{q}(t)\hat{q}(t')) \rangle_\beta = \theta(t - t')D^>(t, t') + \theta(t' - t)D^<(t, t') \quad (2.30)$$

$D^>(t, t')$ and $D^<(t, t')$ depend only on the difference $(t - t')$. This is due to the fact that the propagator must be invariant in time translations and should only depend on the difference in time and not the actual end points. Therefore, we write $D^<(t) = D^<(t, 0)$ and $D^>(t) = D^>(t, 0)$. We define the Fourier transforms of $D^>(t)$ and $D^<(t)$ as [4]

$$D^>(k_0) = \int_{-\infty}^{\infty} dt e^{ik_0 t} D^>(t) \quad (2.31)$$

Similarly,

$$D^<(k_0) = \int_{-\infty}^{\infty} dt e^{ik_0 t} D^<(t) = \int_{-\infty}^{\infty} dt e^{ik_0 t} D^>(t - i\beta) \quad (2.32)$$

Where the second equality in (2.32) arises from the periodicity property. From these we get the relations

$$D^<(k_0) = D^>(-k_0) = e^{-\beta k_0} D^>(k_0) \quad (2.33)$$

The spectral function $\rho(k_0)$ is defined as

$$\rho(k_0) = D^>(k_0) - D^<(k_0) \quad (2.34)$$

From equations (2.26) and (2.27) we get

$$\begin{aligned} \rho(k_0) &= \int_{-\infty}^{\infty} dt e^{ik_0 t} (D^>(t) - D^<(t)) \\ &= \int_{-\infty}^{\infty} dt e^{ik_0 t} (\langle \hat{q}(t)\hat{q}(0) \rangle_\beta - \langle \hat{q}(0)\hat{q}(t) \rangle_\beta) \\ &= \int_{-\infty}^{\infty} dt e^{ik_0 t} \langle [\hat{q}(t), \hat{q}(0)] \rangle_\beta \end{aligned} \quad (2.35)$$

Thus we see that the spectral function is nothing but the Fourier transform of the expectation value of the commutator of $\hat{q}(t)$ at initial and later times.

Equation (2.33) and (2.34) can be solved to get the spectral function in terms of the two point functions in Fourier space.

$$D^>(k_0) = (1 + f(k_0))\rho(k_0) \quad (2.36)$$

$$D^<(k_0) = f(k_0)\rho(k_0) \quad (2.37)$$

where

$$f(k_0) = \frac{1}{e^{\beta k_0} - 1} \quad (2.38)$$

The spectral function also follows a sum rule [4]. We start with the equal time canonical commutation relation

$$[\hat{q}(t), \frac{d}{dt}\hat{q}(t)] = i \quad (2.39)$$

Consider the following integral which can be written as

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{dk_0}{2\pi} k_0 e^{-ik_0 t} \rho(k_0) &= \int_{-\infty}^{\infty} \frac{dk_0}{2\pi} k_0 e^{-ik_0 t} (D^>(k_0) - D^<(k_0)) \\ &= i \frac{d}{dt} (D^>(t) - D^<(t)) \\ &= i \frac{d}{dt} \langle [\hat{q}(t), \hat{q}(0)] \rangle_{\beta} \\ &= i \langle [\frac{d}{dt}\hat{q}(t), \hat{q}(0)] \rangle_{\beta} \end{aligned} \quad (2.40)$$

Taking the limit t tends to zero we get

$$\int_{-\infty}^{\infty} \frac{dk_0}{2\pi} k_0 \rho(k_0) = 1 \quad (2.41)$$

which is the sum rule for the spectral function.

An important case is that of the harmonic oscillator the generalization of which will be

used in the case of fields. We find the free spectral function of a harmonic oscillator using the position operator of the oscillator.

$$\hat{q}(t) = \frac{1}{\sqrt{2\omega}}(ae^{-i\omega t} + a^\dagger e^{i\omega t}) \quad (2.42)$$

We then write using (2.35)

$$D^>(t) - D^<(t) = \frac{1}{2\omega} \langle [a, a^\dagger] e^{-i\omega t} + [a^\dagger, a] e^{i\omega t} \rangle_\beta \quad (2.43)$$

Taking the Fourier transform we get the free spectral function as [4]

$$\rho_F(k_0) = 2\pi\epsilon(k_0)\delta(k_0^2 - \omega^2) \quad (2.44)$$

where $\epsilon(k_0)$ is the sign function.

Thus we see that the free spectral function is temperature independent. In general the spectral function is not temperature independent like. The addition of interactions to the system can introduce temperature dependence. What the spectral function physically means is, it tells us what frequency modes are being excited and how strong. For the free oscillator case, the spectral function is a delta function indicating that the excitations are strongly localized to the natural frequency. Moreover this spectral function can be generalized to scalar fields as scalar fields can be reduced to a continuum of independent harmonic oscillators.

2.4 Matsubara (Imaginary Time) Propagator

We define the Fourier transform of the imaginary time propagator (2.29) as

$$\Delta(i\omega_n) = \int_0^\beta d\tau e^{i\omega_n \tau} \Delta(\tau) \quad (2.45)$$

Since $\Delta(\tau)$ is periodic with a period β , the Fourier transform is taken over the interval $[0, \beta]$. Due to this periodicity, the inverse Fourier transform will be a discrete sum over the variable ω_n which is called the Matsubara [4]. Thus

$$\Delta(\tau) = T \sum_n e^{-i\omega_n \tau} \Delta(i\omega_n) \quad (2.46)$$

where

$$\omega_n = \frac{2\pi n}{\beta} \quad (2.47)$$

Therefore, the addition of temperature to the system causes the imaginary time frequency modes to become discrete. If we choose τ in the interval $[0, \beta]$, from equations (2.29) and (2.31) we have

$$\Delta(\tau) = \int \frac{dk_0}{2\pi} e^{-k_0 \tau} D^>(k_0) \quad (2.48)$$

Substituting in equation (2.42)

$$\Delta(i\omega_n) = \int \frac{dk_0}{2\pi} \int_0^\beta d\tau e^{i\omega_n \tau} e^{-k_0 \tau} D^>(k_0) \quad (2.49)$$

integrating out τ and using equation (2.36), we get the propagator in terms of the spectral function as

$$\Delta(i\omega_n) = - \int \frac{dk_0}{2\pi} \frac{\rho(k_0)}{i\omega_n - k_0} \quad (2.50)$$

$\Delta(i\omega_n)$ is called the Matsubara (full) propagator [4].

2.5 Partition Function of Real Scalar Fields

The Lagrangian density of a scalar field $\phi(x)$ with particle mass m in Minkowski space is given by

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2 \phi^2 - \mathcal{V}(\phi) \quad (2.51)$$

Here $\mathcal{V}(x)$ represents the interactions.

The Euclidean action is written by making time imaginary i.e. $t \rightarrow -i\tau$. Therefore, the derivative with respect to time becomes

$$\frac{\partial}{\partial t} = \frac{\partial}{\partial \tau} \frac{\partial t}{\partial \tau} = i \frac{\partial}{\partial \tau} = i \partial_\tau \quad (2.52)$$

So the Lagrangian becomes

$$\begin{aligned} \mathcal{L} &= -\frac{1}{2}(\partial_\tau \phi)^2 - \frac{1}{2}(\nabla \phi)^2 - \frac{1}{2}m^2 \phi^2 - \mathcal{V}(\phi) \\ &= -[\frac{1}{2}(\partial_\tau \phi)^2 + \frac{1}{2}(\nabla \phi)^2 + \frac{1}{2}m^2 \phi^2 + \mathcal{V}(\phi)] \\ &= \mathcal{L}_E \end{aligned} \quad (2.53)$$

where $\mathcal{L}_E$ is the euclidean Lagrangian. Then the Euclidean action is given by

$$\begin{aligned} S_E[\beta] &= \int_0^\beta d^4x \mathcal{L}_E \\ &= \int_0^\beta d^4x [\frac{1}{2}(\partial_\nu \phi)^2 + \frac{1}{2}m^2 \phi^2 + \mathcal{V}(\phi)] \end{aligned} \quad (2.54)$$

where $(\partial_\nu \phi)^2 = (\partial_\tau \phi)^2 + (\nabla \phi)^2$.

Using this we get the partition function from (2.16) as

$$Z(\beta) = \int \mathcal{D}\phi e^{-S_E[\beta]} \quad (2.55)$$

We write the generating functional as in QFT by introducing source terms[4]

$$Z(\beta; j) = \int \mathcal{D}\phi \exp[-S_E[\beta] + \int_0^\beta d^4x j(x)\phi(x)] \quad (2.56)$$

with $Z(\beta) = Z(\beta; j = 0)$.

The imaginary time integration will run from ∞ to $-\infty$ in the case where $T = 0$ as β becomes infinite which means the system has a period of infinite length. Thus, the result

will reduce to the standard results in QFT.

For the case of a free field, $\mathcal{V}(\phi) = 0$ and the integration above, over ϕ can be converted into standard gaussian form. So we write

$$Z(\beta; j) = \int \mathcal{D}\phi \exp\left[-\left[\frac{1}{2}(\partial_\nu \phi)^2 + \frac{1}{2}m^2 \phi^2\right] + \int_0^\beta d^4x j(x)\phi(x)\right] \quad (2.57)$$

using the identity

$$\begin{aligned} \partial_\nu[\phi \partial_\nu \phi] &= \partial_\nu \phi \partial_\nu \phi + \phi \partial_\nu^2 \phi \\ \partial_\nu \phi \partial_\nu \phi &= \partial_\nu[\phi \partial_\nu \phi] - \phi \partial_\nu^2 \phi = -\phi \partial_\nu^2 \phi \end{aligned} \quad (2.58)$$

where the term $\partial_\nu[\phi \partial_\nu \phi]$ can be converted to a surface integral term becomes zero due to chosen boundary conditions. Substituting back in the above equation for $Z(\beta; j)$

$$\begin{aligned} Z(\beta; j) &= \int \mathcal{D}\phi \exp\left[-\left[-\frac{1}{2}\phi \partial_\nu^2 \phi + \frac{1}{2}m^2 \phi^2\right] + \int_0^\beta d^4x j(x)\phi(x)\right] \\ &= \int \mathcal{D}\phi \exp\left[-\frac{1}{2}\phi[-\partial_\nu^2 + m^2]\phi + \int_0^\beta d^4x j(x)\phi(x)\right] \end{aligned} \quad (2.59)$$

This is in the form of a standard gaussian integral. On evaluating we get

$$Z(\beta; j) = Z(\beta) \exp\left[\frac{1}{2} \int_0^\beta d^4x d^4y j(x)G(x, y)j(y)\right] \quad (2.60)$$

where $G(x, y)$ is the inverse of the operator $(-\partial_\nu^2 + m^2)$ i.e.

$$(-\partial_\nu^2 + m^2)G(x, y) = \delta^4(x - y) \quad (2.61)$$

We now try to find the form and physical meaning of the Greens function $G(x, y)$, to do so we take the functional derivative of (2.57) with respect to the sources [4]

$$\frac{\delta Z(\beta; j)}{\delta j(x)} = Z(\beta)G(x, y)j(y) \exp\left[\frac{1}{2} \int_0^\beta d^4x d^4y j(x)G(x, y)j(y)\right] \quad (2.62)$$

$$\begin{aligned} \frac{\delta^2 Z(\beta; j)}{\delta j(x) \delta j(y)} &= Z(\beta) G(x, y) \exp\left[\frac{1}{2} \int_0^\beta d^4 x d^4 y j(x) G(x, y) j(y)\right] \\ &+ Z(\beta) G^2(x, y) j(x) j(y) \exp\left[\frac{1}{2} \int_0^\beta d^4 x d^4 y j(x) G(x, y) j(y)\right] \end{aligned} \quad (2.63)$$

Setting source to zero i.e. $j = 0$

$$\frac{1}{Z(\beta)} \frac{\delta^2 Z(\beta; j)}{\delta j(x) \delta j(y)} \Big|_{j=0} = G(x, y) \quad (2.64)$$

Similarly we take the functional derivative of (2.53). Since $S_E(\beta)$ is independent of source terms, only the second term in the exponent will contribute. Thus we get

$$\frac{1}{Z(\beta)} \frac{\delta^2 Z(\beta; j)}{\delta j(x) \delta j(y)} \Big|_{j=0} = \int \mathcal{D}\phi \phi(x) \phi(y) e^{-S_E[\beta]} \quad (2.65)$$

The LHS in the above equation is nothing but the time ordered expectation $\langle T(\phi(x) \phi(y)) \rangle_\beta$ in imaginary time $-i\tau$ which can be derived in the same manner as for the partition function in path integral form from $\langle T(\phi(x) \phi(y)) \rangle_\beta = \text{Tr}(e^{-\beta \hat{H}} \hat{\phi}(x) \hat{\phi}(y))$.

From (2.61), (2.62) and generalizing (2.24) to fields by including spatial coordinates as well for the argument we get

$$G(x, y) = \int \mathcal{D}\phi \phi(x) \phi(y) e^{-S_E[\beta]} = \langle T(\phi(x) \phi(y)) \rangle_\beta = \Delta_F(x - y) \quad (2.66)$$

where F denotes free field case.

Thus the Greens function $G(x, y)$ is nothing but the position space thermal propagator. It gives the probability amplitude that the field will go from an initial configuration to a final one. Therefore we can write

$$(-\partial_\nu^2 + m^2) \Delta_F(x - y) = \delta^4(x - y) \quad (2.67)$$

To solve this equation in accordance to the periodicity in β we rewrite the equation in

Fourier space [4]

$$\begin{aligned}(\omega_n^2 + k^2 + m^2)\Delta_F(i\omega_n, k) &= 1 \\ (\omega_n^2 + \omega_k^2)\Delta_F(i\omega_n, k) &= 1\end{aligned}\tag{2.68}$$

where

$$\begin{aligned}\omega_k^2 &= k^2 + m^2 \quad (\text{On shell condition}) \\ \omega_n &= \frac{2\pi n}{\beta} \quad (\text{Matsubara frequencies})\end{aligned}$$

and k represents momentum. Therefore, we get the propagator in Fourier Space or the Matsubara propagator

$$\Delta_F(i\omega_n, k) = \frac{1}{\omega_n^2 + \omega_k^2}\tag{2.69}$$

By taking Fourier transform with respect to imaginary time, we get a mixed representation in imaginary time τ and momentum k

$$\begin{aligned}\Delta_F(\tau, k) &= T \sum_n e^{-i\omega_n \tau} \frac{1}{\omega_n^2 + \omega_k^2} \\ &= \frac{1}{2\omega_k} ((1 + n(\omega_k)e^{-\omega_k \tau} + n(\omega_k)e^{\omega_k \tau})\end{aligned}\tag{2.70}$$

where $n(\omega)$ is the Bose-Einstein distribution

$$n(\omega) = \frac{1}{e^{\beta\omega} - 1}$$

Thus we naturally get back the distribution for spinless scalar particles (bosons).

The propagator in position space is obtained by another Fourier transform for momentum

$$\Delta_F(x) = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{x}} \Delta_F(\tau, k)\tag{2.71}$$

Note that, due to translational symmetry we have replaced $(x - y)$ with just x in the

argument i.e. setting $y = 0$

2.6 Conclusion

We see in this chapter that the partition function can be obtained by treating the statistical weight $e^{-\beta\hat{H}}$ as an evolution in imaginary time and can be expressed in a path integral form. We defined the real time and imaginary time (Matsubara) propagator and studied the spectral function for the case of a free harmonic oscillator. We ended by extending the formulation to field and derived the partition function and the propagator for a real scalar field.

Chapter 3

Real Time Formalism

3.1 Introduction

Till now we have restricted time to imaginary. But dynamics in the real world happen in real time. So to take this into account we, instead of restricting time to be imaginary, promote it to being complex and the time integration is performed along a contour in this complex time plane. The contour chosen must contain the real axis as we are interested in real time dynamics [4].

3.2 The CTP Contour

We know that we can write the expectation value of any operator A as

$$\langle A \rangle = Tr[\rho(t)A] \tag{3.1}$$

where $\rho(t)$ represents the density matrix. Since the density matrix can be written as

$$\rho(t) = \frac{U(t,0)e^{-\beta\hat{H}}U(0,t)}{Tr[e^{-\beta\hat{H}}]} \tag{3.2}$$

where $U(t, 0)$ is the evolution operator, we get

$$\langle A \rangle = \frac{\text{Tr}[U(t, 0)e^{-\beta\hat{H}}U(0, t)A]}{\text{Tr}[e^{-\beta\hat{H}}]} \quad (3.3)$$

Using the fact that the term $e^{-\beta\hat{H}}$ can be thought of as an evolution operator in imaginary time, we get [3]

$$\begin{aligned} \langle A \rangle &= \frac{\text{Tr}[U(t, 0)U(T - i\beta, T)U(0, t)A]}{\text{Tr}[U(T - i\beta, T)]} \\ &= \frac{\text{Tr}[U(T - i\beta, T)U(0, t)AU(t, 0)U(0, T)U(T, 0)]}{\text{Tr}[U(T - i\beta, T)]} \\ &= \frac{\text{Tr}[U(T - i\beta, T)U(T, 0)U(0, t)AU(t, 0)U(0, T)]}{\text{Tr}[U(T - i\beta, T)]} \\ &= \frac{\text{Tr}[U(T - i\beta, T)U(T, t)AU(t, T)]}{\text{Tr}[U(T - i\beta, T)]} \\ &= \frac{\text{Tr}[U(T - i\beta, T)U(T, T')U(T', T)U(T, t)AU(t, T)]}{\text{Tr}[U(T - i\beta, T)U(T, T')U(T', T)]} \\ &= \frac{\text{Tr}[U(T - i\beta, T)U(T, T')U(T', t)AU(t, T)]}{\text{Tr}[U(T - i\beta, T)U(T, T')U(T', T)]} \end{aligned} \quad (3.4)$$

This shows us that the evolution of the system goes from a time T to t where the operator A is applied and then evolves to time T' and back to T . Finally the system evolves from T to $T - i\beta$ along imaginary time. Now T and T' can be taken to be $-\infty$ and ∞ respectively. This will define our contour in the complex plane.

3.3 Definition of the Two-Point Function

We wish to calculate the two point Green's function

$$G_C(x, y) = \langle T_C(\hat{\phi}(x), \hat{\phi}(y)) \rangle_\beta \quad (3.5)$$

where the subscript C represents integration along the contour and T_C denotes that the ordering in time is done in the complex plane along a chosen contour. This time ordering

along the contour can be defined by writing the contour in a parametric form. We define $t = z(u)$ where the parameter u is real and strictly increasing. So a ordering in complex time is mapped to an ordering in the parameter u [4].

For this formalism, we redefine the θ and δ functions as[4]

$$\theta_C(t - t') = \theta(u - u'), \quad \delta_C(t - t') = \left(\frac{\partial z}{\partial u}\right)^{-1} \delta(u - u') \quad (3.6)$$

We also redefine functional differentiation as

$$\frac{\delta j(x)}{\delta j(y)} = \delta_C(t - t') \delta^3(x - y) \quad (3.7)$$

where the function $j(x)$ lives on the contour C

The propagator in (2.30) can be generalized as

$$D_C(x - y) = \theta_C(t - t') D_C^>(x, y) + \theta_C(t' - t) D_C^<(x, y) \quad (3.8)$$

A useful representation of the propagator is given by [4]

$$D_C(x - y) = \int \frac{d^4 k}{(2\pi)^4} e^{-ik(x-y)} [\theta_C(t - t') + f(k_0)] \rho(k) \quad (3.9)$$

where $\rho(k)$ is the spectral function generalized to four dimensions and $f(k_0)$ is

$$f(k_0) = \frac{1}{e^{\beta k_0} - 1} \quad (3.10)$$

and (3.9) has the symmetry property [4]

$$D_C(x - y) = D_C(x' - y) \quad (3.11)$$

3.4 Closed Time Path

The contour is restricted to start from an initial time t_i and should end at a $t_i - i\beta$. This is a direct consequence arising from the KMS condition. Along with this the imaginary part of t must be non-increasing in the parameter u [4]. The contour must also contain the real axis. A standard choice satisfying these criteria is as follows [4]

1. C starts from t_i which is real (later taken to be $-\infty$)
2. C goes up till $-t_i$ and this segment is labeled as C_1
3. The contour then goes from $-t_i$ to $-t_i - i\sigma$ along the imaginary axis where $0 \leq \sigma \leq \beta$ (this segment is labeled as C_3)
4. The path then goes back from $-t_i - i\sigma$ to $t_i - i\sigma$ labeled by C_2
5. Lastly the path goes from $t_i - i\sigma$ to $t_i - i\beta$ similar to C_3 and is called C_4

We know that the free generating functional for a free scalar field is given by

$$Z_C(\beta; j) = \mathcal{N} \exp\left[-\frac{1}{2} \int_C d^4x d^4y j(x) D_C(x-y) j(y)\right] \quad (3.12)$$

where $\mathcal{N}$ is the normalization constant and C denotes that the time integration is done along the contour C in complex plane. According to [4] the generating functional can be factorized into two separate horizontal and vertical branch components when t_i goes to $-\infty$

$$Z_C(\beta; j) = \mathcal{N} Z_{C_{12}}(\beta; j) Z_{C_{34}}(\beta; j) \quad (3.13)$$

where C_{12} means the integration is done along paths C_1 and C_2 (horizontal branch), similarly for C_{34} (vertical branch). This means that the integration along the vertical branch amounts to a constant and can be absorbed into the normalization constant by redefining

it. Thus the generating functional is rewritten as

$$Z_C(\beta; j) = \mathcal{N}' \exp\left[-\frac{1}{2} \int_{C_{12}} d^4x d^4y j(x) D_C(x-y) j(y)\right] \quad (3.14)$$

where $\mathcal{N}$ is the redefined normalization constant.

Since the time integration now runs between ∞ and $-\infty$ on two different branches i.e. C_1 and C_2 we label the source $j(x)$ by the branch they exist on as

$$j_1 = j(t, \mathbf{x}) \quad j_2 = j(t - i\sigma, \mathbf{x}) \quad (3.15)$$

With this (3.14) becomes

$$Z_C(\beta; j) = \mathcal{N}' \exp\left[-\frac{1}{2} \int_{-\infty}^{\infty} d^4x d^4y j_a(x) D_{ab}(x-y) j_b(y)\right] \quad (3.16)$$

Thus we get a propagator involving 4 components as (denoting only using the time argument)

$$\begin{aligned} D_{11} &= D(t - t') \\ D_{22} &= D^*(t - t') \\ D_{12} &= D_C(t - (t' - i\sigma)) \\ D_{21} &= D_C((t - i\sigma) - t') \end{aligned} \quad (3.17)$$

When (3.16) is written in its equivalent path integral form, it becomes

$$Z_C(\beta; j) = \int \mathcal{D}\phi_1 \mathcal{D}\phi_2 \exp\left[-\frac{1}{2} \int_{-\infty}^{\infty} d^4x d^4y \phi_a(x) K_{ab} \phi_b(y) + i \int_{-\infty}^{\infty} d^4x j_a(x) \phi_a(x)\right] \quad (3.18)$$

where K_{ab} is the inverse of $D_{ab}(x-y)$. This equation when simplified by performing the standard gaussian integration gives back (3.16).

By carrying out functional differentiation of (3.18) with respect to sources we find that

the matrix elements of the propagator matrix are time ordered thermal expectation of the field operators i.e.

$$D_{ab}(x, y) = \langle T(\phi_a(x), \phi_b(y)) \rangle_\beta \quad (3.19)$$

where the subscripts a,b represent the time branch the field operator lives in and the time ordering is done according to the contour parameter v . Thus all times on the second time branch lies in the future of those on the first branch.

The explicit expressions for the propagators in (3.17) can be found from the expression (3.9) by choosing the appropriate contour path. The expressions have been given in [4] and will not be needing them as we will be expressing our results in terms of the spectral function defined in the preceding chapter.

3.5 Conclusion

Since we are interested in real time dynamics, we studied a formulation different from the imaginary time formalism. The real time formalism includes the real time dynamics by making the time variable a complex number and performing integrations along a contour in the complex plane. We discussed why this formalism makes sense and derived the propagator accordingly. We see that the propagator comes out to be a 2x2 matrix whose components are the time ordered thermal expectations of the field operators on different time branches of the contour. This formalism will be used to study the open system dynamics in the next chapter.

Chapter 4

Open Quantum Systems

Here we look into the dynamics of quantum systems interacting with an environment. We will take our system to be a real scalar field interacting with a second scalar field which will be taken as the environment, and derive the dynamics using the Feynman-Vernon influence functional method [2]. We will then extend the general interaction with the environment to the case where the environment is in thermal equilibrium with T as its temperature and will behave as a bath for our system.

4.1 Closed and Open Systems

A closed quantum system evolves according to its own internal unitary Hamiltonian dynamics and does not interact with any external system. The system is effectively isolated.

On the other hand a quantum system is considered to be open if it interacts with its surroundings affecting its dynamics. Due to this nature of the system interacting with its surroundings, it can be thought as a quantum system S (dynamics of which we are interested in) coupled to a second quantum system E called the environment (we are not interested in how this system behaves)[1].

Therefore the system S is a subsystem of the combined system $S + E$ which is taken to be isolated. Since the combined system plus environment system is isolated, it evolves

according to Hamiltonian dynamics[1].

Due to the interaction between the system S and the environment, the net changes in the state of the system S is caused by its internal dynamics and that induced by the environment, cannot be represented using unitary evolution.

The dynamics of S induced by the evolution of the total system is called reduced system dynamics and S is called the reduced system[1].

Let $\mathcal{H}_S$ and $\mathcal{H}_E$ represent the respective Hilbert spaces of the system and the environment. Then the Hilbert space of the combined system is then the tensor product of the individual Hilbert spaces.

$$\mathcal{H}_{total} = \mathcal{H}_S \otimes \mathcal{H}_E \quad (4.1)$$

Using this, the total Hamiltonian of the system can be written as

$$H(t) = H_S \otimes I_E + I_S \otimes H_E + \hat{H}_I(t) \quad (4.2)$$

where H_S and H_E are the free Hamiltonian of the systems S and E respectively and $\hat{H}_I(t)$ represents the Hamiltonian which describes the interactions between the two subsystems. The interaction Hamiltonian may depend on time. I_S and I_E are identity operators on the respective Hilbert spaces of S and E [1].

It is assumed that all the observables we are interested in, refer to the subsystem S . All the observables of the total system which refers to S are of the form

$$A \otimes I_E$$

where A is an observable in the Hilbert space of S .

Let ρ represent the density matrix of the total system, then the expectation value of an

observable A is given by

$$\begin{aligned}\langle A \rangle &= Tr_E[(A \otimes I_E)\rho] \\ &= Tr[A\rho_S]\end{aligned}\tag{4.3}$$

where Tr_E denotes partial trace over the environment and $\rho_S = Tr_E[\rho]$ is the reduced density matrix for the subsystem S .

We know that the evolution of the density matrix from an initial time t_0 to a final time t is given by [1]

$$\rho(t) = U(t, t_0)\rho(t_0)U^\dagger(t, t_0)\tag{4.4}$$

Using this, the evolution of the reduced density matrix is obtained by taking partial trace on both sides. Thus

$$\rho_S(t) = Tr_E[U(t, t_0)\rho(t_0)U^\dagger(t, t_0)]\tag{4.5}$$

The equation of motion for the density matrix is given by the von Neumann equation

$$\frac{d}{dt}\rho(t) = -i[H(t), \rho(t)]\tag{4.6}$$

The equation of motion for the reduced density matrix can then be obtained in a similar way by taking the partial trace of the von Neumann equation. Thus

$$\frac{d}{dt}\rho_S(t) = -iTr_E[[H(t), \rho(t)]]\tag{4.7}$$

4.2 Open System Dynamics of Fields

In many realistic situations, the system we are interested in, is not isolated but interacts with an external environment. To describe such scenarios, we divide the total degrees of freedom into a system field ϕ and an environment field ψ . In the path integral formulation which we will be using, the tracing procedure explained in the above section leads naturally

to the concept of the influence functional, which encodes the effect of the environment on the system [2]. We will take the case where both ϕ and ψ are free real scalar fields and assume that ϕ is the system field whose dynamics we are interested in and ψ is the environment it interacts with.

The free action integral for the scalar fields are obtained from their respective Lagrangian densities

$$S[\phi] = \int d^4x \left(\frac{1}{2} \partial_\mu \phi(x) \partial^\mu \phi(x) - \frac{1}{2} m^2 \phi^2(x) \right) \quad (4.8)$$

$$S[\psi] = \int d^4x \left(\frac{1}{2} \partial_\mu \psi(x) \partial^\mu \psi(x) - \frac{1}{2} M^2 \psi^2(x) \right) \quad (4.9)$$

where m and M are the masses of their respective particles. The interaction between both the fields in general is modeled as a polynomial coupling of the form[2]

$$S_I[\phi, \psi] = \int d^4x \lambda f[\phi(x)] \psi^k(x)$$

where $f[\phi(x)]$ is a function of the field operator $\phi(x)$. The constant lambda determines the strength of the interactions between both fields is assumed to be small. Here we take the case of a simple form of interaction where $k = 1$ and $f[\phi(x)] = \phi(x)$ and S_I will become

$$S_I[\phi, \psi] = \int d^4x \lambda \phi(x) \psi(x) \quad (4.10)$$

Thus the interaction action becomes bilinear in $\phi(x)$ and $\psi(x)$

Thus the full action of the combined system is written as

$$S_{total}[\phi, \psi] = S[\phi] + S[\psi] + S_I[\phi, \psi] \quad (4.11)$$

The density matrix representing the entire system is defined as [2]

$$\rho[\phi^1, \psi^1, \phi^2, \psi^2, t] = \langle \phi^1, \psi^1 | \hat{\rho}(t) | \phi^2, \psi^2 \rangle \quad (4.12)$$

where the density matrix is written in the eigen basis of the field operators $\hat{\phi}(x)$ and $\hat{\psi}(x)$. The superscripts of the 1,2 will represent the different real time integration branches in the real time formalism. We will see later that the integral we obtain has the same form as the one we obtained for the partition function in the previous chapter and will be used to derive the propagator. Inserting a complete set of eigen states at $t = t_i$

$$\begin{aligned}\rho[\phi^1, \psi^1, \phi^2, \psi^2, t] &= \int d\phi_i^1 d\phi_i^2 d\psi_i^1 d\psi_i^2 \langle \phi^1, \psi^1 | \phi_i^1 \psi_i^1 \rangle \langle \phi_i^1 \psi_i^1 | \hat{\rho}(t_i) | \phi_i^2 \psi_i^2 \rangle \langle \phi_i^2 \psi_i^2 | \phi^2, \psi^2 \rangle \\ &= \int d\phi_i^1 d\phi_i^2 d\psi_i^1 d\psi_i^2 \int_{\phi_i^1, \psi_i^1}^{\phi^1, \psi^1} \mathcal{D}\phi^1 \mathcal{D}\psi^1 e^{\frac{iS[\phi^1, \psi^1]}{\hbar}} \rho[\phi_i^1, \psi_i^1, \phi_i^2, \psi_i^2, t_i] \\ &\quad \int_{\phi_i^2, \psi_i^2}^{\phi^2, \psi^2} \mathcal{D}\phi^2 \mathcal{D}\psi^2 e^{\frac{-iS[\phi^2, \psi^2]}{\hbar}}\end{aligned}\quad (4.13)$$

where we have replaced $\langle \phi^1, \psi^1 | \phi_i^1 \psi_i^1 \rangle$ and $\langle \phi_i^2 \psi_i^2 | \phi^2, \psi^2 \rangle$ with their respective equivalent path integral expression and $S[\phi, \psi]$ is the total action as in (4.11).

This equation describes the evolution of the total density matrix which can be rewritten in terms of an evolution operator as

$$\rho[\phi^1, \psi^1, \phi^2, \psi^2, t] = \int d\phi_i^1 d\phi_i^2 d\psi_i^1 d\psi_i^2 \mathcal{J}[\phi_f^1, \psi_f^1, \phi_f^2, \psi_f^2, t | \phi_f^1, \psi_f^1, \phi_i^2, \psi_i^2, t_i] \rho[\phi_i^1, \psi_i^1, \phi_i^2, \psi_i^2, t_i]$$

where $\mathcal{J}[\phi_f^1, \psi_f^1, \phi_f^2, \psi_f^2, t | \phi_f^1, \psi_f^1, \phi_i^2, \psi_i^2, t_i]$ is the evolution operator. We assume that initially the state was uncorrelated. This means that before the initial time $t = t_i$ both the system field states were independent of each other and the interaction was turned on and allowed to interact from the time $t = t_i$ onwards. Since the respective field were independent, the total density matrix of the combined system will become the tensor product of their individual density matrices i.e. $\rho_{total} = \rho_S \otimes \rho_E[1]$. Thus the total density matrix (4.12) can be factorized as

$$\rho[\phi_i^1, \psi_i^1, \phi_i^2, \psi_i^2, t_i] = \rho_\phi[t_i] \times \rho_\psi[t_i] \quad (4.14)$$

where we have written $\rho_\phi[t_i]$ and $\rho_\psi[t_i]$ instead of $\rho[\phi_i^1, \phi_i^2, t_i]$ and $\rho[\psi_i^1, \psi_i^2, t_i]$ respectively for simplicity[2]. Substituting in (4.13) we get

$$\rho[t] = \int d\phi_i^1 d\phi_i^2 d\psi_i^1 d\psi_i^2 \int_{\phi_i^1, \psi_i^1}^{\phi^1, \psi^1} \mathcal{D}\phi^1 \mathcal{D}\psi^1 e^{\frac{iS[\phi^1, \psi^1]}{\hbar}} \rho_\phi[t_i] \rho_\psi[t_i] \int_{\phi_i^2, \psi_i^2}^{\phi^2, \psi^2} \mathcal{D}\phi^2 \mathcal{D}\psi^2 e^{\frac{-iS[\phi^2, \psi^2]}{\hbar}} \quad (4.15)$$

We can now find the reduced system dynamics by tracing out the environment field $\psi(x)$ as explained before. This is done by integrating over the final configuration of the environment field $\psi(x)$ as

$$\rho_r[\phi^1, \phi^2, t] = \int d\psi_f \rho[t] \quad (4.16)$$

which is equivalent to taking the partial trace of the matrix in (4.3)[2]. Therefore we get

$$\begin{aligned} \rho_r[\phi^1, \phi^2, t] = \int d\psi_f d\phi_i^1 d\phi_i^2 d\psi_i^1 d\psi_i^2 \int_{\phi_i^1, \psi_i^1}^{\phi^1, \psi^1} \mathcal{D}\phi^1 \mathcal{D}\psi^1 \rho_\phi[t_i] \rho_\psi[t_i] \\ \int_{\phi_i^2, \psi_i^2}^{\phi^2, \psi^2} \mathcal{D}\phi^2 \mathcal{D}\psi^2 e^{\frac{i}{\hbar}(S[\phi^1, \psi^1] - S[\phi^2, \psi^2])} \end{aligned} \quad (4.17)$$

Thus we get the full evolution equation of the system field S whose dynamics we are interested in. Focusing on the term in the exponent

$$\begin{aligned} S[\phi^1, \psi^1] - S[\phi^2, \psi^2] &= (S[\phi^1] + S[\psi^1] + S_I[\phi^1, \psi^1]) - (S[\phi^2] + S[\psi^2] + S_I[\phi^2, \psi^2]) \\ &= (S[\phi^1] - S[\phi^2]) + (S[\psi^1] - S[\psi^2] + S_I[\phi^1, \psi^1] - S_I[\phi^2, \psi^2]) \\ &= (S[\phi^1] - S[\phi^2]) + S' \end{aligned} \quad (4.18)$$

where $S' = (S[\psi^1] - S[\psi^2] + S_I[\phi^1, \psi^1] - S_I[\phi^2, \psi^2])$ is the component of the action which depend on the environment field $\psi(x)$. Substituting back we get

$$\begin{aligned} \rho_r[\phi^1, \phi^2, t] &= \int \mathcal{D} e^{\frac{i}{\hbar}(S[\phi^1] - S[\phi^2] + S')} \\ &= \int \mathcal{D} e^{\frac{i}{\hbar}(S[\phi^1] - S[\phi^2])} e^{\frac{i}{\hbar} S'} \end{aligned} \quad (4.19)$$

where we have written the integration measure including the density matrix function as $\int \mathcal{D}$ for simplicity.

We now define the functional

$$\begin{aligned}\mathcal{F}[\phi^1, \phi^2] &= e^{\frac{i}{\hbar} S_{IF}[\phi^1, \phi^2]} \\ &= \int d\psi_f d\psi_i^1 \psi_i^2 \rho_\psi[t_i] \int_{\psi_i^1}^{\psi^1} \mathcal{D}\psi^1 \int_{\psi_i^2}^{\psi^2} \mathcal{D}\psi^2 e^{\frac{i}{\hbar} S'}\end{aligned}\quad (4.20)$$

This functional is called the influence functional and S_{IF} is called the influence action[2][1].

We define

$$S_{eff} = S[\phi^1] - S[\phi^2] + S_{IF}[\phi^1, \phi^2] \quad (4.21)$$

Thus, (4.19) becomes

$$\begin{aligned}\rho_r[\phi^1, \phi^2, t] &= \int d\phi_i^1 \phi_i^2 \int_{\phi_i^1}^{\phi^1} \mathcal{D}\phi^1 \int_{\phi_i^2}^{\phi^2} \mathcal{D}\phi^2 e^{\frac{i}{\hbar} S_{eff}} \rho_\phi[t_i] \\ &= \int d\phi_i^1 \phi_i^2 \mathcal{J}_r[\phi_f^1, \phi_f^2 | \phi_i^1, \phi_i^2] \rho_\phi[t_i]\end{aligned}\quad (4.22)$$

where

$$\mathcal{J}_r[\phi_f^1, \phi_f^2 | \phi_i^1, \phi_i^2] = \int_{\phi_i^1}^{\phi^1} \mathcal{D}\phi^1 \int_{\phi_i^2}^{\phi^2} \mathcal{D}\phi^2 e^{\frac{i}{\hbar} S_{eff}} \quad (4.23)$$

can be thought of as the evolution operator of the reduced system. Focusing on the path integral of the influence functional in (4.20)

$$\int_{\psi_i^1}^{\psi^1} \mathcal{D}\psi^1 \int_{\psi_i^2}^{\psi^2} \mathcal{D}\psi^2 e^{\frac{i}{\hbar} S'} = \int_{\psi_i^1}^{\psi^1} \mathcal{D}\psi^1 \int_{\psi_i^2}^{\psi^2} \mathcal{D}\psi^2 e^{\frac{i}{\hbar} ((S[\psi^1] - S[\psi^2] + S_I[\phi^1, \psi^1] - S_I[\phi^2, \psi^2]))} \quad (4.24)$$

we look at the free action $S[\psi^1]$. From (4.8)

$$S[\psi^1] = \int d^4x \left(\frac{1}{2} \partial_\mu \phi(x) \partial^\mu \phi(x) - \frac{1}{2} m^2 \phi^2(x) \right) \quad (4.25)$$

on integrating by parts we get

$$\begin{aligned} S[\psi^1] &= -\frac{1}{2} \int d^4x \psi^1 (\partial_\nu \partial^\nu + M^2) \psi^1 \\ &= -\frac{1}{2} \int d^4x \psi^1 K \psi^1 \end{aligned} \quad (4.26)$$

where $K = (\partial_\nu \partial^\nu + M^2)$. Similarly for $S[\psi^2]$

$$S[\psi^2] = -\frac{1}{2} \int d^4x \psi^2 K \psi^2 \quad (4.27)$$

Therefore, S' on substitution

$$S' = \int d^4x \left[-\frac{1}{2} \psi^1 K \psi^1 + \lambda \phi^1 \psi^1 + \frac{1}{2} \psi^2 K \psi^2 - \lambda \phi^2 \psi^2 \right] \quad (4.28)$$

Let $\lambda \phi^1 = j^1$ and $\lambda \phi^2 = j^2$. Then

$$S' = \int d^4x \left[-\frac{1}{2} \psi^1 K \psi^1 + j^1 \psi^1 + \frac{1}{2} \psi^2 K \psi^2 - j^2 \psi^2 \right] \quad (4.29)$$

We see that here $\phi(x)$ behaves as a source for the action S' . Let

$$\Psi = \begin{bmatrix} \psi^1 \\ \psi^2 \end{bmatrix}, \quad J = \begin{bmatrix} j^1 \\ -j^2 \end{bmatrix}, \quad \mathcal{K} = \begin{bmatrix} K & 0 \\ 0 & -K \end{bmatrix} \quad (4.30)$$

Then (4.29) can be written in matrix form as

$$S' = \int d^4x \left[-\frac{1}{2} \Psi^a \mathcal{K}_{ab} \Psi^b + J^a \Psi^a \right] \quad (4.31)$$

substituting in (4.24), the resulting integral is gaussian

$$\int_{\psi_i^1}^{\psi^1} \mathcal{D}\psi^1 \int_{\psi_i^2}^{\psi^2} \mathcal{D}\psi^2 e^{\frac{i}{\hbar} S'} = \int_{\psi_i^1}^{\psi^1} \mathcal{D}\psi^1 \int_{\psi_i^2}^{\psi^2} \mathcal{D}\psi^2 \exp\left[\frac{i}{\hbar} \left(\int d^4x \left[-\frac{1}{2} \Psi^a \mathcal{K}_{ab} \Psi^b + J^a \Psi^a \right] \right) \right] \quad (4.32)$$

Using standard result of gaussian integrals, we get

$$\int_{\psi_i^1}^{\psi^1} \mathcal{D}\psi^1 \int_{\psi_i^2}^{\psi^2} \mathcal{D}\psi^2 e^{\frac{i}{\hbar} S'} = (\det \mathcal{K})^{-\frac{1}{2}} \exp\left[\frac{i}{2\hbar} \int d^4x d^4y J^a(x) \mathcal{G}_{ab}(x-y) J^b(y)\right] \quad (4.33)$$

where $\mathcal{G}_{ab}(x-y)$ is the inverse of $\mathcal{K}_{ab}$ i.e.

$$\mathcal{K} \mathcal{G}(x-y) = \delta^4(x-y) \quad (4.34)$$

The coefficient $(\det \mathcal{K})^{-\frac{1}{2}}$ diverges, but since its a constant and does not contribute to the dynamics of the system, we can drop it as it can be removed by renormalization. Thus

$$e^{\frac{i}{\hbar} S_{IF}} = \exp\left[\frac{i\lambda^2}{2\hbar} \int d^4x d^4y \phi^a(x) \mathcal{G}_{ab}(x-y) \phi^b(y)\right] \quad (4.35)$$

From functional differentiation as done in the previous chapters and taking into consideration the contour time ordering we get that the, matrix $\mathcal{G}_{ab}(x-y)$ has the components[2]

$$\mathcal{G}_{ab}(x-y) = \frac{i\lambda^2}{\hbar} \begin{bmatrix} \langle T(\psi(x)\psi(y)) \rangle & -\langle \psi(y)\psi(x) \rangle \\ -\langle \psi(x)\psi(y) \rangle & \langle \tilde{T}(\psi(y)\psi(x)) \rangle \end{bmatrix} \quad (4.36)$$

where T and $\tilde{T}$ represents time and anti time ordering respectively

Using this matrix form we can expand S_{IF} as

$$\begin{aligned} S_{IF} = \frac{i\lambda^2}{2\hbar} \int d^4x d^4y [& \langle T(\psi(x)\psi(y)) \rangle \phi^1(x) \phi^1(y) - \langle \psi(y)\psi(x) \rangle \phi^1(x) \phi^2(y) \\ & - \langle \psi(x)\psi(y) \rangle \phi^2(x) \phi^1(y) + \langle \tilde{T}(\psi(y)\psi(x)) \rangle \phi^2(x) \phi^2(y)] \end{aligned} \quad (4.37)$$

We have found the equation for the influence action. To bring out the physics from the equation we make the substitution $\phi^1(x) = \phi_c(x) + \frac{\phi_q(x)}{2}$ and $\phi^2(x) = \phi_c(x) - \frac{\phi_q(x)}{2}$. We get

$$S_{IF} = \int d^4x d^4y [\phi_q(x) \mathbf{D}(x,y) \phi_c(y) + \frac{i}{2} \phi_q(x) \mathbf{N}(x,y) \phi_q(y)] \quad (4.38)$$

where

$$\mathbf{D}(x, y) = \frac{i\lambda^2}{\hbar} \langle [\psi(x), \psi(y)] \rangle \theta(t - t'), \quad \mathbf{N}(x, y) = \frac{\lambda^2}{2\hbar} \langle \{\psi(x), \psi(y)\} \rangle \quad (4.39)$$

Here $\mathbf{D}(x, y)$ is called the dissipation and $\mathbf{N}(x, y)$ is called the noise kernel [2]. The dissipation kernel corresponds to the retarded propagator.

Before the substitution, $\phi^1(x)$ and $\phi^2(x)$ represented the copies of the field in the forward and backward time branched. The substitution we used above is called the Keldysh basis and realizes physical phenomenon as $\phi_c(x)$ represents the physical field in the classical limit and $\phi_q(x)$ takes into account the quantum noises in the field.

4.3 Connection with Thermal Field Theory

If our environment field $\psi(x)$ is in thermal equilibrium at temperature T , the field operator expectations in the matrix elements of $\mathcal{G}_{ab}(x-y)$ will be replaced by their respective thermal averages instead which will lead to the expectations in the dissipation and noise kernels to be replaced by their thermal variants as well. Thus introducing temperature into the formalism. The dissipation and noise kernels then become

$$\mathbf{D}(x, y) = \frac{i\lambda^2}{\hbar} \langle [\psi(x), \psi(y)] \rangle_\beta \theta(t - t'), \quad \mathbf{N}(x, y) = \frac{\lambda^2}{2\hbar} \langle \{\psi(x), \psi(y)\} \rangle_\beta \quad (4.40)$$

Generalizing (2.26), (2.27), (2.36) and (2.37) to fields in 4 dimensions, we can write the thermal expectations after using inverse Fourier transform as

$$\langle [\psi(x), \psi(y)] \rangle_\beta = \int_{-\infty}^{\infty} \frac{d^4 k}{(2\pi)^4} e^{-ik(x-y)} \rho(k) \quad (4.41)$$

and

$$\langle \{\psi(x), \psi(y)\} \rangle_\beta = \int_{-\infty}^{\infty} \frac{d^4 k}{(2\pi)^4} e^{-ik(x-y)} (1 + 2f(k_0)) \rho(k) \quad (4.42)$$

Generalizing the free spectral function (2.44), the dissipation and noise kernel becomes

$$\begin{aligned} \mathbf{D}(x, y) &= \frac{i\lambda^2}{\hbar} \theta(t - t') \int_{-\infty}^{\infty} \frac{d^4 k}{(2\pi)^4} e^{-ik(x-y)} \rho_F(k) \\ \mathbf{N}(x, y) &= \frac{\lambda^2}{2\hbar} \int_{-\infty}^{\infty} \frac{d^4 k}{(2\pi)^4} e^{-ik(x-y)} (1 + 2f(k_0)) \rho_F(k) \end{aligned} \quad (4.43)$$

Where the generalized free spectral function is given by

$$\rho_F(k) = 2\pi\epsilon(k_0)\delta(k^2 - m^2) \quad (4.44)$$

Since the spectral function of a free field is independent of temperature, the dissipation kernel is also temperature dependent. The dependency on temperature shows up in the noise kernel through the term $f(k_0)$. This means that the thermal interactions create strong temperature dependent noise in the system field $\phi(x)$.

From (4.40) we can see that since the dissipator kernel is imaginary, it gives rise to an exponentially decaying term in the influence functional, thus showing that system is not in equilibrium and dissipates into the environment field. Similarly the noise kernel is real and gives rise to an oscillatory term in the influence functional encoding the fluctuations in the system. So to Sum it up the real part of the influence functional gives rise to dissipative behavior whereas the noise is given by the imaginary part.

It is also important to notice that, both the dissipation and noise kernels are non local in nature even in time. However causality is maintained by the the theta function which selects the correct time ordering. This structure has a clear physical interpretation. The non-local dependence on spacetime coordinates indicates that the system's evolution depends on its past history, reflecting memory effects induced by the environment. Systems like these, which have memory are said to be Non-Markovian in nature. Thus, our example model of two interacting fields follows Non-Markovian dynamics.

4.4 Conclusion

We studied the behavior of two scalar fields interacting with each other via a bilinear coupling by deriving the influence functional of the field. We then found the corresponding non local dissipation and noise kernels and saw that the dissipation part is temperature independent and temperature enters the picture via the noise kernel. Due to the non-local nature of the kernels, the system has memory which mean the dynamics depends on past events. Thus we decribed the open dynamics for our chosen system.

Chapter 5

Conclusion

In this report we started with the formalism for the thermalization of quantum fields using imaginary time where we saw that the statistical weight term $e^{-\beta\hat{H}}$ can be interpreted as an evolution operator in imaginary time. Due to this we see that the field must satisfy periodic conditions in imaginary time with a period of β and the imaginary time axis can be thought of being compactified into a circle of circumference β . We then derived the partition function and propagators from first principles.

We then studied the real time formalism which allows us to describe the thermal dynamics in real time domain as real world phenomena take place in real time and not imaginary. This led to the 2x2 propagator matrix form where the elements are nothing but the different time ordered correlation functions.

In the last chapter we discussed the the dynamics when the field is kept open and interacting with the environment. Using the method of influence functional we derived its dynamics and found the dissipation and noise kernels for two free scalar fields coupled together.

It was noticed that the kernels were non-local in nature indicating the system had memory and hence was non-Markovian in nature.

We then added temperature to the environment field by keeping it in thermal equilib-

rium with a heat bath. This lead to only the noise kernel having a temperature dependence. In a general case were the fields are not free and have interaction terms, the dissipation energy will also have temperature dependencies. This dissipation kernel leads to a decaying exponential term in the evolution equation implying that the system is open and losing energy, whereas the noise kernel contributes an oscillatory term which encodes all the quantum noises occurring in the system.

Thus, integrating out the environmental degrees of freedom yields an effective, non-local description of the system that incorporates both dissipation and fluctuations, providing a natural bridge between thermal field theory and open quantum system dynamics.

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